

WASP-36b

R-1498

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_210226 · created 2026-08-01T14:31:35+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459271.8088 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.037
Transit depth (Rp/Rs)^2	2.15 +/- 1.08 [%]
Orbital Inclination (inc)	88.1 +/- 2.05
Airmass coefficient 1 (a1)	1.128 +/- 0.025
Airmass coefficient 2 (a2)	-0.102 +/- 0.037
Scatter in the residuals of the lightcurve fit is	2.18 %
Best Comparison Star	#2 - [353, 159]
Optimal Method	PSF photometry
Transit Duration (day)	0.0856 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.147 ± 0.037 vs 0.1368 (deviation 0.27σ)
Duration fit vs published	0.0856 d vs 0.0758 d (deviation 2.18σ)
Mid-time O–C	-7.9 min
Depth SNR	3.9
Residual scatter	2.18 %

Light curve

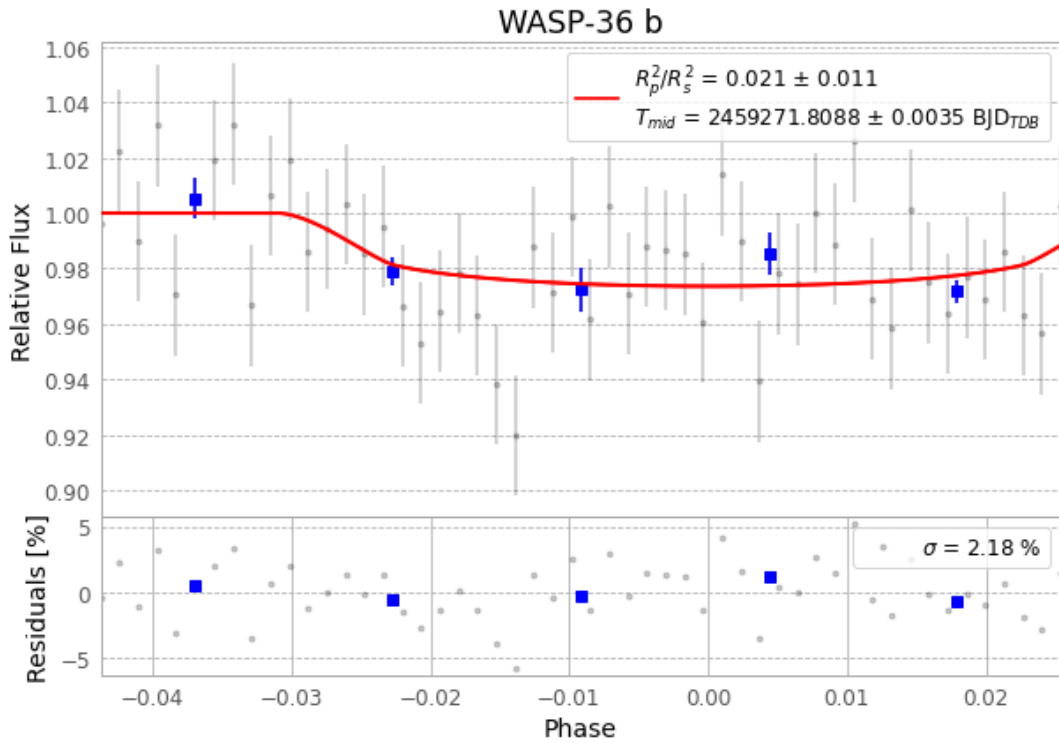


Figure 1 — FinalLightCurve_WASP-36 b_2021-02-25.png (SHA-256 ab2a345c82349dbd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [438, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 baf6770b0254235c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

52 FITS frames (34 MB), WASP-36210226053912.FITS → WASP-36210226081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-210226.log (b57d0a8ab42a...), FinalLightCurve_WASP-36 b_2021-02-25.png (ab2a345c8234...), FinalLightCurve_WASP-36 b_2021-02-25.csv (aea32f737c9c...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:31Z.